

A Study of Factors Influencing House Prices

Group number and group members

Group member Contribution:

- Alice (EDA and modeling)
- Bob (model diagnostics and model comparisons)
- Carol (conclusion and report preparation)

Abstract

This project investigates the relationship between various house features and their sale prices using a subset of the Ames Housing dataset, which includes the records of 2874 houses in Ames, Iowa. We perform exploratory data analysis, fit linear models with the full set and a subset of variables, with and without interactions, and also apply two shrinkage methods. We carry out necessary diagnostics via numerical and graphical tools and adopt a few remedial strategies, including a log transformation to the sale price before linear modeling. We compare a series of models using F-test, AIC, and cross-validation. The final model we choose balances simplicity and model fit; it explains 89.62% of the variation in log(Sale Price) and highlights a few significant factors, such as the overall quality, total area, year built or remodeled, as well as the zoning classification. There also exists significant interaction between zoning classification and lot area. Finally, to ensure the validity of the conclusion in this report, further examination on a few outliers may be helpful.

Exploratory Data Analysis

This data has the following sample size and number of predictors:

```
paste("sample size:", nrow(data))

## [1] "sample size: 2930"

paste("number of predictors:", ncol(data) - 1)

## [1] "number of predictors: 14"
```

In particular, the variables included in this dataset are summarized as follows:

##	Sale_Price	Overall_Qual	Gr_Liv_Area	Garage_Cars	
##	Min. : 12789	Length:2930	Min. : 334	Min. :0.000	
##	1st Qu.:129500	Class :character	1st Qu.:1126	1st Qu.:1.000	
##	Median :160000	Mode :character	Median :1442	Median :2.000	
##	Mean :180796		Mean :1500	Mean :1.766	
##	3rd Qu.:213500		3rd Qu.:1743	3rd Qu.:2.000	
##	Max. :755000		Max. :5642	Max. :5.000	
##	Garage_Area	Total_Bsmt_SF	Full_Bath	Year_Built	Year_Remod_Add
##	Min. : 0.0	Min. : 0	Min. :0.000	Min. :1872	Min. :1950
##	1st Qu.: 320.0	1st Qu.: 793	1st Qu.:1.000	1st Qu.:1954	1st Qu.:1965
##	Median : 480.0	Median : 990	Median :2.000	Median :1973	Median :1993
##	Mean : 472.7	Mean :1051	Mean :1.567	Mean :1971	Mean :1984

```
## 3rd Qu.: 576.0    3rd Qu.:1302    3rd Qu.:2.000    3rd Qu.:2001    3rd Qu.:2004
## Max.    :1488.0    Max.    :6110    Max.    :4.000    Max.    :2010    Max.    :2010
##      Lot_Area      Fireplaces      Garage_Type      Mo_Sold
## Min.    : 1300    Min.    :0.0000    Length:2930      Min.    : 1.000
## 1st Qu.: 7440    1st Qu.:0.0000    Class :character    1st Qu.: 4.000
## Median : 9436    Median :1.0000    Mode  :character    Median : 6.000
## Mean   : 10148    Mean   :0.5993      Mean   : 6.216
## 3rd Qu.: 11555    3rd Qu.:1.0000      3rd Qu.: 8.000
## Max.   :215245    Max.   :4.0000      Max.   :12.000
##      Year_Sold      MS_Zoning
## Min.    :2006    Length:2930
## 1st Qu.:2007    Class :character
## Median :2008    Mode  :character
## Mean   :2008
## 3rd Qu.:2009
## Max.   :2010
```

We note that there are three variables not identified as numerical: Overall_Qual, MS_Zoning and Garage. The values and distributions of MS_Zoning are as follows:

```
##
##              A_agr              C_all
##              2              25
## Floating_Village_Residential      I_all
##              139              2
##      Residential_High_Density      Residential_Low_Density
##              27              2273
##      Residential_Medium_Density
##              462
```

There are few houses with Zoning code A_agr, C_all, I_all, and Residential_High_Density. We will disregard these levels and focus on levels with more samples, in order to get more accurate results.

The values and distributions of Garage_type are as follows:

```
##
##      Attchd      Basment      BuiltIn      CarPort
##      1720      34      186      13
##      Detchd More_Than_Two_Types      No_Garage
##      757      22      142
```

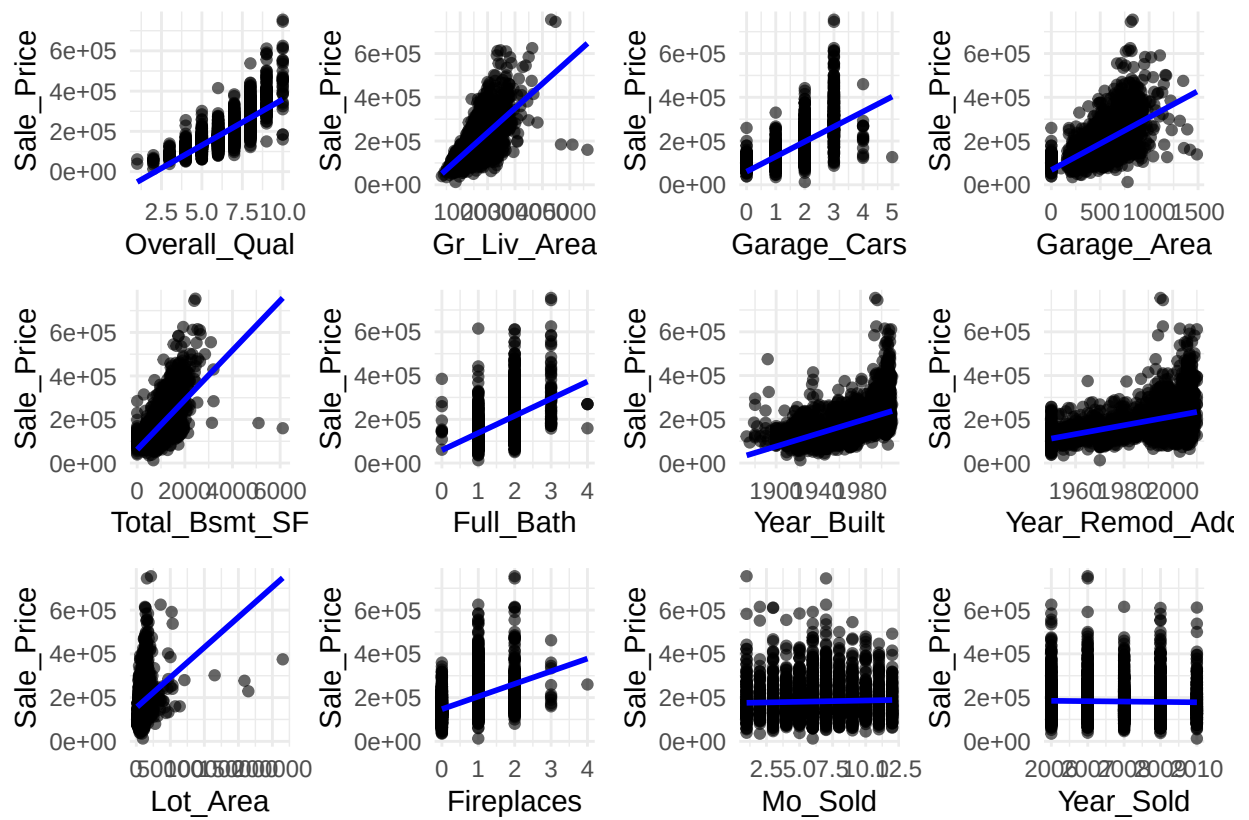
There may be too many types of Garages, making the analysis too complicated. In addition, the biggest difference may lie between having at least one garage vs. not having any ([other ways of combining levels may also be appropriate](#)).

While for the Overall_Qual, we notice that it is an ordinal variable with inherent ordering between categories:

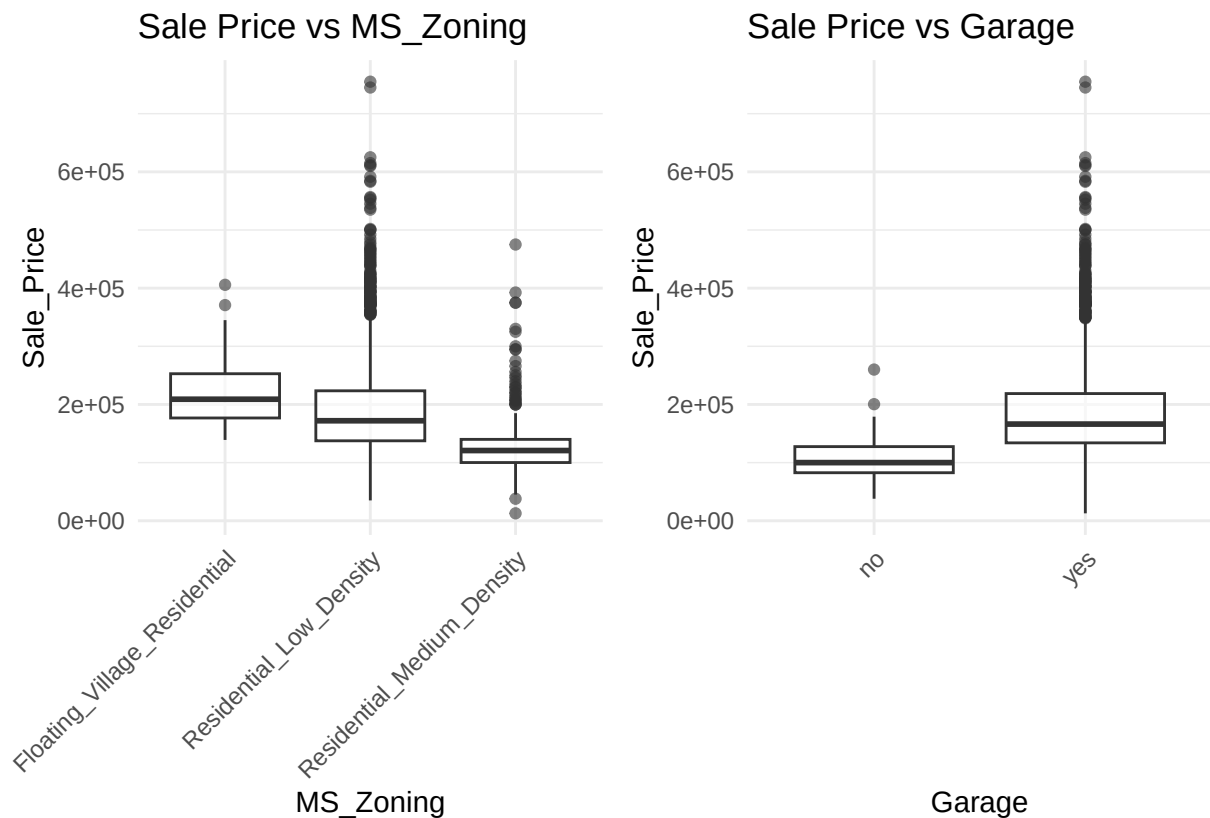
```
##
## Above_Average      Average Below_Average      Excellent      Fair
##      720      810      210      107      33
##      Good      Poor Very_Excellent      Very_Good      Very_Poor
##      600      11      31      350      2
```

We decide to transform it into numerical variables ranging from 1 to 10, where “Very_Poor” means a score of 1, and “Very_Excellent” leads to a score of 10.

After these preprocessing, scatter plots between the Sale_Price and all numerical variables are as follows (to save space, you can include only a subset of scatter plots where you see strong patterns):



Observations: Variables like Overall_Qual, Gr_Liv_Area, Total_Bsmt_SF show strong correlation with Sale_Price. A few outliers seem to exist.



Ob-

servations: There seem to be some group difference amongst different levels of `Garage` and `MS_Zoning`.

Model Building

Linear model with all variables, without interactions (Model 1)

We first fit a linear model with all the predictors, and compare it with the intercept-only model:

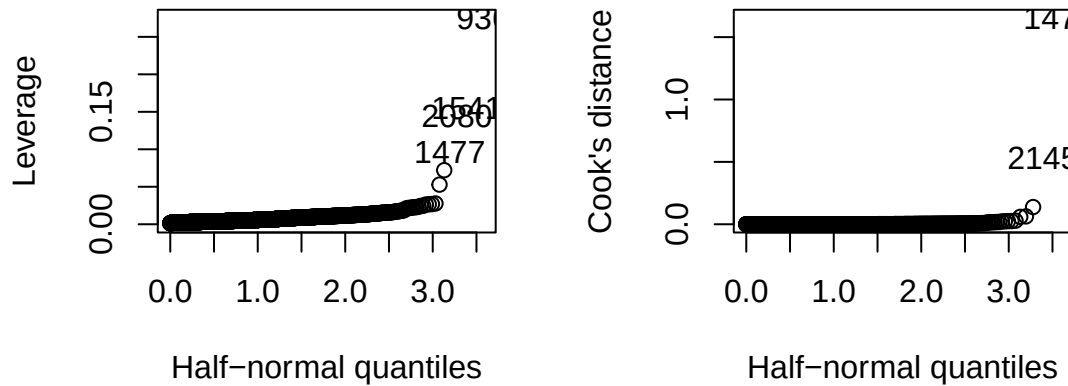
```
##
## Call:
## lm(formula = Sale_Price ~ ., data = data)
##
## Coefficients:
##              (Intercept)              Overall_Qual
##              1.864e+05              1.997e+04
##              Gr_Liv_Area              Garage_Cars
##              4.647e+01              7.243e+03
##              Garage_Area              Total_Bsmt_SF
##              3.826e+01              2.626e+01
##              Full_Bath              Year_Built
##              -7.178e+03              2.251e+02
##              Year_Remod_Add              Lot_Area
##              3.474e+02              5.655e-01
##              Fireplaces              Mo_Sold
##              9.288e+03              -6.483e+01
##              Year_Sold              MS_ZoningResidential_Low_Density
##              -6.834e+02              4.673e+03
## MS_ZoningResidential_Medium_Density              Garageyes
##              -2.872e+03              -2.028e+04

## Analysis of Variance Table
##
## Model 1: Sale_Price ~ 1
## Model 2: Sale_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Cars + Garage_Area +
##          Total_Bsmt_SF + Full_Bath + Year_Built + Year_Remod_Add +
##          Lot_Area + Fireplaces + Mo_Sold + Year_Sold + MS_Zoning +
##          Garage
##   Res.Df      RSS Df Sum of Sq    F    Pr(>F)
## 1    2873 1.8262e+13
## 2    2858 3.5943e+12 15 1.4668e+13 777.53 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Finding: It seems that the model explains significant variation in `Sale_Price`.

Diagnostics for the basic linear model

We first check unusual observations.



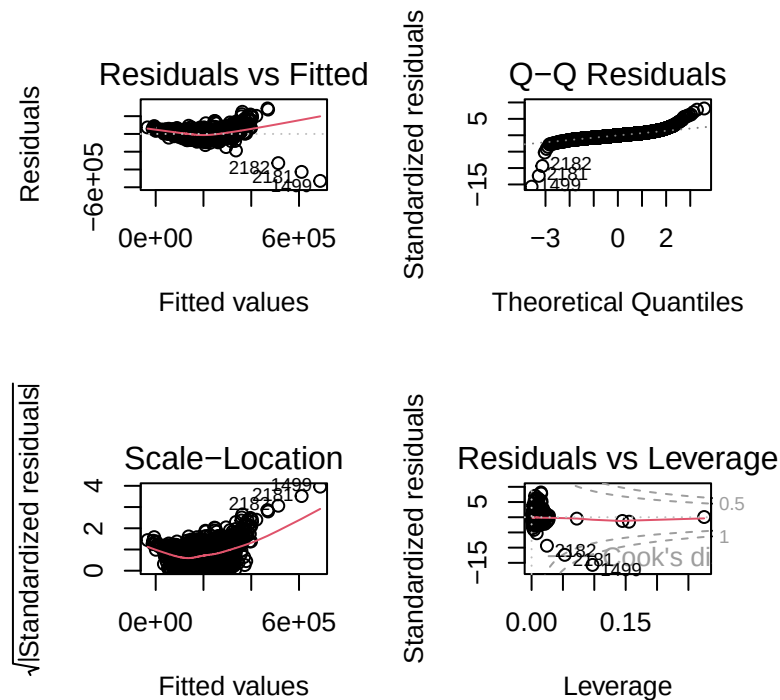
Findings: The halfnormal plots above suggest some high leverage points and highly influential point. By the rule of thumb, there are 223 high leverage points, and one highly influential point (sample number 1477). We may need to check how the model changes if removing this sample.

We also check the outliers by performing an outlier test:

```
## 16 45 424 433 434 457 1064 1183 1499 1638 1761 1768 2181 2182 2331 2333
## 15 43 414 423 424 447 1042 1161 1477 1607 1727 1734 2145 2146 2288 2290
## 2335 2446 2451
## 2292 2402 2407
```

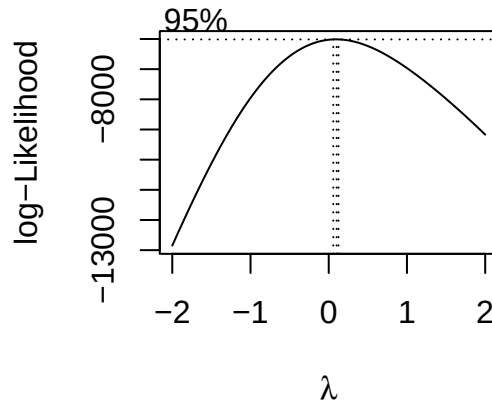
Findings: We find 19 outliers.

The following are diagnostic plots for the linear model with all predictors:



Findings: Some non-normality in residuals and slight heteroscedasticity observed. The plots also suggests non-linearity and outliers.

Remedy: Motivated by the diagnostics, we consider Box Cox transformation.



We round the optimal λ to the nearest common values in Box Cox transformation: let $\lambda = 0$. That is, we use log transformation.

Refit a linear model with log(Price) as the response (Model 2)

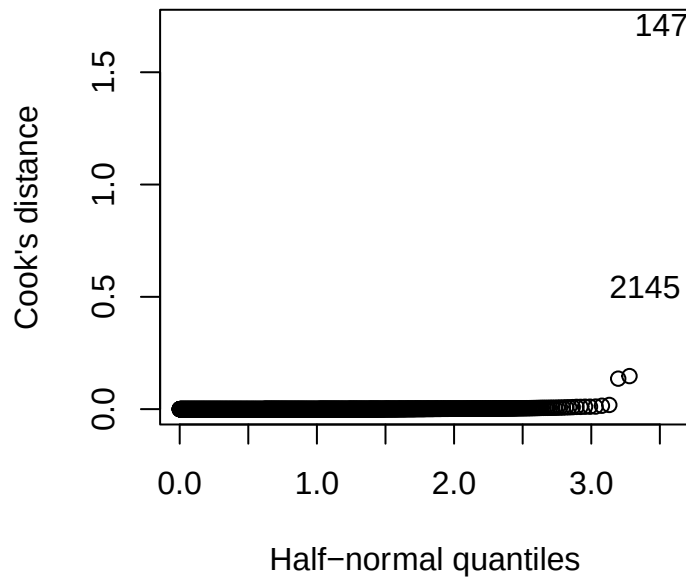
We still use all the predictors to fit a linear model, no interaction included, and the response variable is log(Sale_Price).

```
##
## Call:
## lm(formula = log_Price ~ . - Sale_Price, data = data)
##
## Coefficients:
##              (Intercept)              Overall_Qual
##              1.122e+01              9.838e-02
##              Gr_Liv_Area              Garage_Cars
##              2.040e-04              4.080e-02
##              Garage_Area              Total_Bsmt_SF
##              7.145e-05              1.105e-04
##              Full_Bath              Year_Built
##              -1.118e-02              1.299e-03
##              Year_Remod_Add              Lot_Area
##              2.589e-03              2.608e-06
##              Fireplaces              Mo_Sold
##              5.707e-02              6.271e-05
##              Year_Sold              MS_ZoningResidential_Low_Density
##              -4.040e-03              3.193e-03
## MS_ZoningResidential_Medium_Density              Garageeyes
##              -9.890e-02              6.199e-02
```

Diagnostics for Model 2

We first check highly influential points and outliers:

```
cook = cooks.distance(mod2)
halfnorm(cook, ylab="Cook's distance")
```



```
which.max(cook)
```

```
## 1499
```

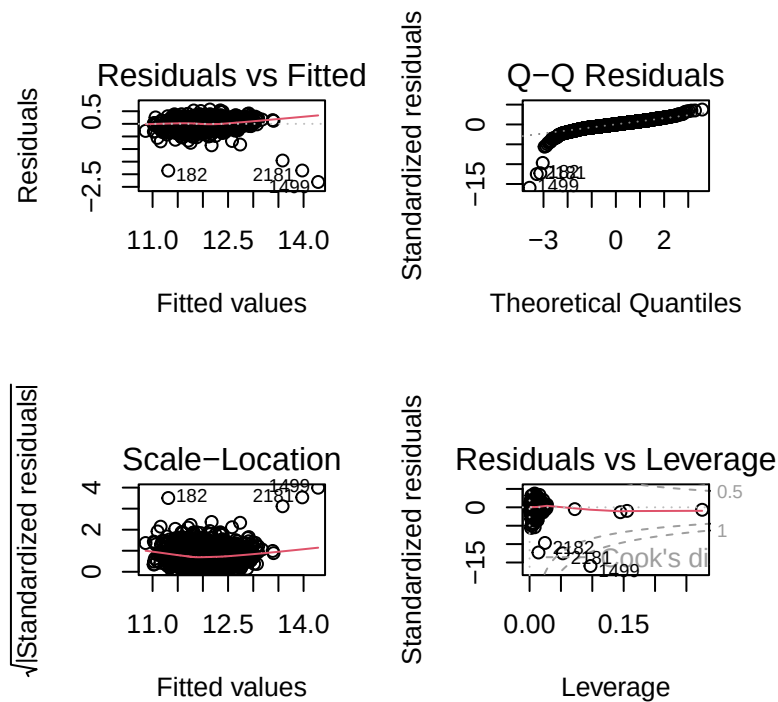
```
## 1477
```

```
## 182 373 562 791 1183 1499 1783 2181 2182 2844
```

```
## 179 363 550 775 1161 1477 1749 2145 2146 2796
```

Findings: There is still one highly influential data point: sample number 1477, and some outliers.

Diagnostic plots for Model 2 are as follows:



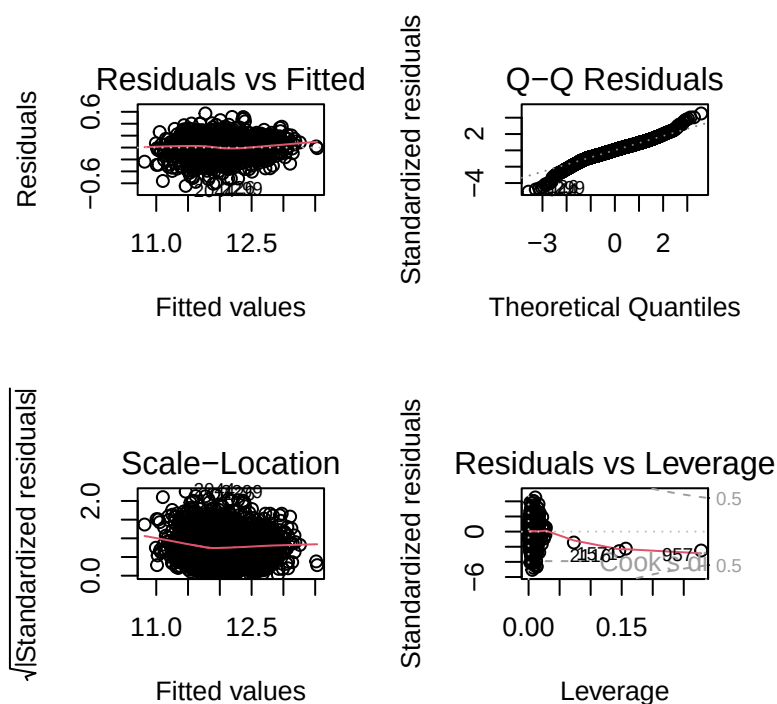
Findings: Aside from the outliers (including the highly influential point), the diagnostic plots look good. We now remove the outliers and refit the model.

Model 3: Remove Outliers

Removing 10 significant outliers, we refit the model with $\log(\text{Sale_Price})$ as the response to get Model 3:

```
##
## Call:
## lm(formula = log_Price ~ ., data = newdata)
##
## Coefficients:
##              (Intercept)              Overall_Qual
##              9.269e+00              8.894e-02
##              Gr_Liv_Area              Garage_Cars
##              2.494e-04              9.600e-03
##              Garage_Area              Total_Bsmt_SF
##              1.508e-04              1.587e-04
##              Full_Bath              Year_Built
##              -2.997e-02              1.501e-03
##              Year_Remod_Add              Lot_Area
##              2.628e-03              3.259e-06
##              Fireplaces              Mo_Sold
##              5.255e-02              -8.054e-05
##              Year_Sold              MS_ZoningResidential_Low_Density
##              -3.314e-03              -5.641e-03
## MS_ZoningResidential_Medium_Density              Garageyes
##              -9.119e-02              7.450e-02
```

```
par(mfrow=c(2,2))
plot(mod3)
```



Findings: Model 3 has slight deviation from normality. Other than this, there does not seem other serious issues.

Variable Selection (Model Comparison with Model 3 Included)

We start from Model 3, and use stepwise procedure to select a subset of predictors, using BIC as the criterion.

```
library(leaps)
step(mod3, direction="both", k=log(nrow(newdata)), trace = 0)

##
## Call:
## lm(formula = log_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Area +
##     Total_Bsmt_SF + Full_Bath + Year_Built + Year_Remod_Add +
##     Lot_Area + Fireplaces + MS_Zoning + Garage, data = newdata)
##
## Coefficients:
##              (Intercept)              Overall_Qual
##              2.568e+00              8.937e-02
##              Gr_Liv_Area              Garage_Area
##              2.501e-04              1.745e-04
##              Total_Bsmt_SF              Full_Bath
##              1.581e-04              -2.888e-02
##              Year_Built              Year_Remod_Add
##              1.535e-03              2.615e-03
##              Lot_Area              Fireplaces
##              3.275e-06              5.279e-02
## MS_ZoningResidential_Low_Density MS_ZoningResidential_Medium_Density
##              -3.924e-03              -8.872e-02
##              Garageyes
##              7.925e-02
```

Findings: The set of selected variables are: Overall_Qual, Gr_Liv_Area, Garage_Area, Total_Bsmt_SF, Full_Bath, Year_Built, Year_Remod_Add, Lot_Area, Fireplaces, MS_Zoning, Garage. We let this model be the Model 4. Based on the BIC criterion, we prefer Model 4 over Model 3.

```
mod4 <- lm(log_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Area +
  Total_Bsmt_SF + Full_Bath + Year_Built + Year_Remod_Add +
  Lot_Area + Fireplaces + MS_Zoning + Garage, data = newdata)
```

ANCOVA Model with Interactions (Model Comparison with Model 4 Included)

We also want to consider a model with interactions between categorical and numerical variables. An F-test that compares the additive model (Model 4) and the interaction model (Model 5) is as follows:

```
## Analysis of Variance Table
##
## Model 1: log_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Area + Total_Bsmt_SF +
##     Full_Bath + Year_Built + Year_Remod_Add + Lot_Area + Fireplaces +
##     MS_Zoning + Garage
## Model 2: log_Price ~ (Overall_Qual + Gr_Liv_Area + Garage_Area + Total_Bsmt_SF +
##     Full_Bath + Year_Built + Year_Remod_Add + Lot_Area + Fireplaces) *
##     (MS_Zoning + Garage)
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    2851 46.423
## 2    2825 44.768 26    1.6544 4.0152 4.133e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Findings: At least one interaction term is significant.

We want to select a subset of interactions for parsimony. So we apply the stepwise selection method again, with Model 4 being the smallest model and Model 5 set as the biggest model. The following code gives us a selected model that is between Model 4 and Model 5:

```
step(mod4, scope = list(upper = mod5), direction = "both", k = log(n), trace = 0)
```

```
##
## Call:
## lm(formula = log_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Area +
##     Total_Bsmt_SF + Full_Bath + Year_Built + Year_Remod_Add +
##     Lot_Area + Fireplaces + MS_Zoning + Garage + Lot_Area:MS_Zoning,
##     data = newdata)
##
## Coefficients:
##                (Intercept)
##                2.166e+00
##            Overall_Qual
##                8.883e-02
##            Gr_Liv_Area
##                2.485e-04
##            Garage_Area
##                1.622e-04
##        Total_Bsmt_SF
##                1.560e-04
##            Full_Bath
##               -3.149e-02
##            Year_Built
##                1.755e-03
##        Year_Remod_Add
##                2.575e-03
##            Lot_Area
##                1.246e-05
##            Fireplaces
##                5.421e-02
##    MS_ZoningResidential_Low_Density
##                6.158e-02
##    MS_ZoningResidential_Medium_Density
##               -8.488e-02
##            Garageyes
##                8.004e-02
##    Lot_Area:MS_ZoningResidential_Low_Density
##               -9.444e-06
##    Lot_Area:MS_ZoningResidential_Medium_Density
##                5.381e-07
```

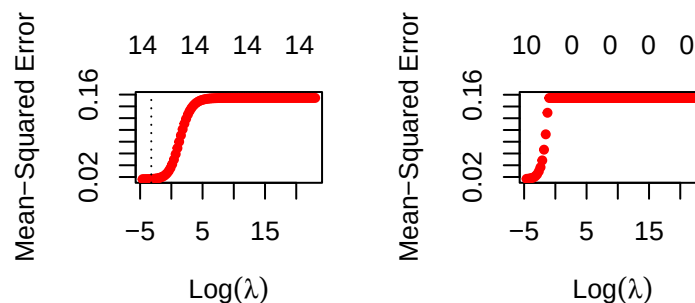
Finding: Only the interaction between Lot_Area and MS_Zoning is kept. Based on the BIC criterion, we choose the following model (Model 6) over the prior ones.

```
##
## Call:
## lm(formula = log_Price ~ Overall_Qual + Gr_Liv_Area + Garage_Area +
##     Total_Bsmt_SF + Full_Bath + Year_Built + Year_Remod_Add +
##     Lot_Area + Fireplaces + MS_Zoning + Garage + Lot_Area:MS_Zoning,
##     data = newdata)
##
## Coefficients:
```

```
##              (Intercept)
##              2.166e+00
##              Overall_Qual
##              8.883e-02
##              Gr_Liv_Area
##              2.485e-04
##              Garage_Area
##              1.622e-04
##              Total_Bsmt_SF
##              1.560e-04
##              Full_Bath
##              -3.149e-02
##              Year_Built
##              1.755e-03
##              Year_Remod_Add
##              2.575e-03
##              Lot_Area
##              1.246e-05
##              Fireplaces
##              5.421e-02
##              MS_ZoningResidential_Low_Density
##              6.158e-02
##              MS_ZoningResidential_Medium_Density
##              -8.488e-02
##              Garageeyes
##              8.004e-02
##              Lot_Area:MS_ZoningResidential_Low_Density
##              -9.444e-06
##              Lot_Area:MS_ZoningResidential_Medium_Density
##              5.381e-07
```

Shrinkage Methods (Ridge and Lasso Regression)

We also try some shrinkage methods including the ridge and Lasso regression. We consider the same set of variables as Model 6, and $\log(\text{Sale_Price})$ as the response. The following figures show the cross-validation errors of the two methods with a range of λ values.



The minimum cross-validation errors for ridge and Lasso regression are:

```
paste("Minimum cross-validation error for Ridge:", min(ridge_mod$cvm))
```

```
## [1] "Minimum cross-validation error for Ridge: 0.0163100193653278"
```

```
paste("Minimum cross-validation error for Lasso:", min(lasso_mod$cvm))
```

```
## [1] "Minimum cross-validation error for Lasso: 0.0162775445508516"
```

The optimal λ for ridge regression is:

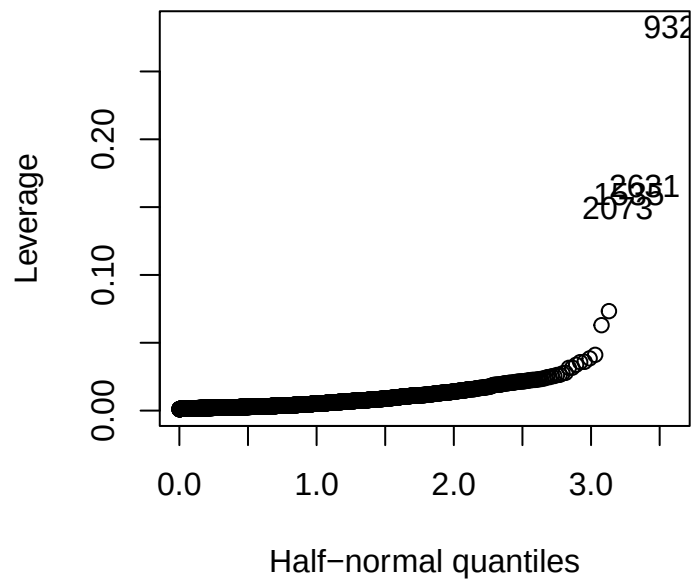
```
ridge_mod$lambda.min
```

```
## [1] 0
```

Finding: Ridge regression with $\lambda = 0$ gives the smallest cross-validation error. This is the same as Model 6. Therefore, the final model selection is Model 6. Before concluding, we do one more diagnostic for Model 6.

Check for high-leverage points:

```
lev=influence(mod6)$hat
p <- ncol(newdata); n <- nrow(newdata)
halfnorm(lev, 4, ylab="Leverage")
```



```
sum(lev>2*p/n)
```

```
## [1] 281
```

281 high leverage points based on the rule of thumb

Check for outliers:

```
jack=rstudent(mod4);
n <- nrow(model.matrix(mod4))
p <- ncol(model.matrix(mod4))
cutoff <- abs(qt(.05/(2*n), n-p-1)) # Bonferroni correction
which(abs(jack) > cutoff)
```

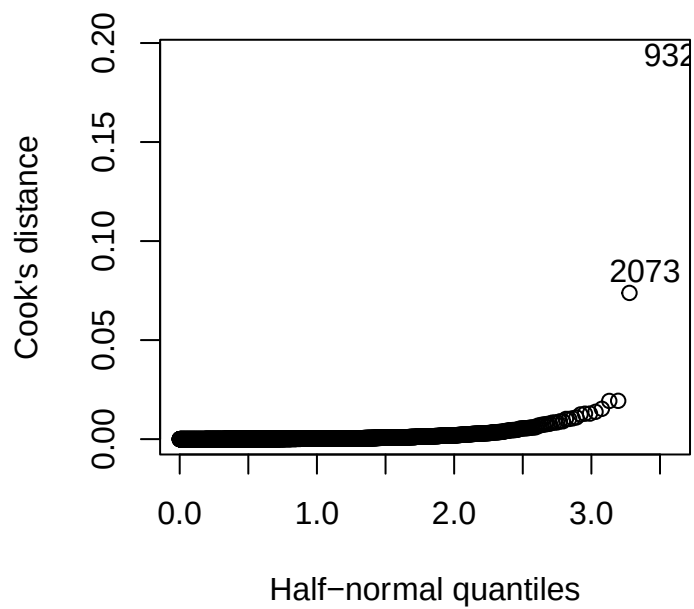
```
## 126 710 754 2044 2299 2893
```

```
## 123 695 736 2001 2247 2828
```

Findings: 6 outliers. Need to report.

Check highly-influential data points:

```
cook = cooks.distance(mod4)
halfnorm(cook, ylab="Cook's distance")
```



```
sort(cook, decreasing = TRUE)[1:5]
```

```
##          957          2116          1571          126          754
## 0.19388355 0.08484876 0.07382282 0.01934535 0.01928305
```

```
which.max(cook)
```

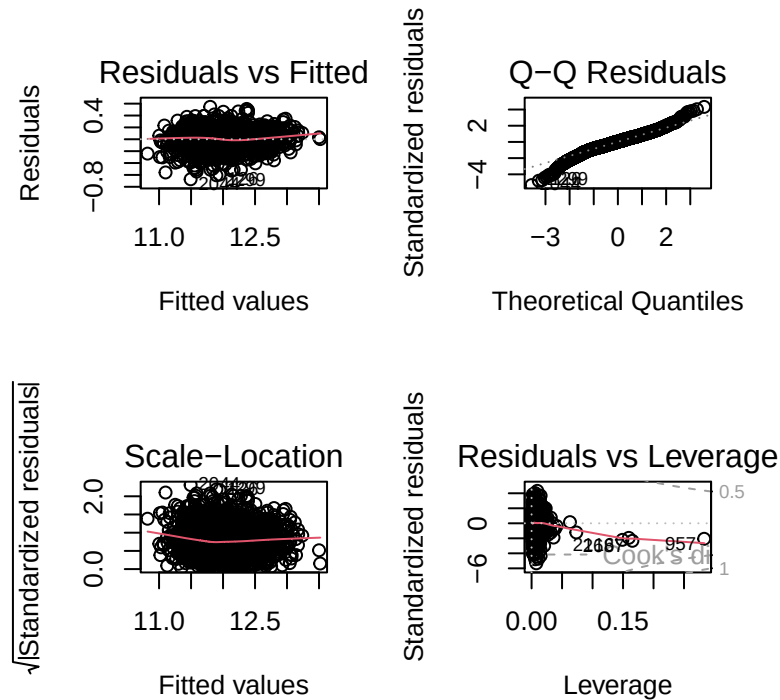
```
## 957
```

```
## 932
```

Findings: No highly influential point.

Diagnostic plots:

```
par(mfrow=c(2,2))
plot(mod6)
```



Findings: Aside from the 6 outliers, there does not seem any serious issues ([You may remove it and check how the model changes, or report it in your conclusion](#)).

Discussion of Results and Conclusions

Based on a series of model building, diagnostics and comparisons, we find that **the following variables are important in determining the sale price of a house:**

- Overall_Quality (Rates of overall material and finish of the house)
- Gr_Liv_Area (Above grade (ground) living area in square feet)
- Garage Area
- Total_Bsmt_SF (Total square feet of basement area)
- Full_Bath (Number of full bathrooms above ground)
- Year Built
- Year Remodeled
- Lot_Area
- number of Fireplaces
- zoning classification
- whether garage is included

There is also **significant interaction between the zoning classification and the area of lot.**

The log of the sale price appears to have a linear relationship with these variables. Specifically, the expectation of the log(sale price) is estimated as a linear combination of these variables, with the following coefficients:

```
coef(mod6)
```

```
##              (Intercept)
##              2.165911e+00
##              Overall_Qual
##              8.882558e-02
##              Gr_Liv_Area
##              2.484517e-04
##              Garage_Area
##              1.621983e-04
##              Total_Bsmt_SF
##              1.560043e-04
##              Full_Bath
##              -3.149348e-02
##              Year_Built
##              1.755163e-03
##              Year_Remod_Add
##              2.575272e-03
##              Lot_Area
##              1.246454e-05
##              Fireplaces
##              5.420771e-02
##              MS_ZoningResidential_Low_Density
##              6.158328e-02
##              MS_ZoningResidential_Medium_Density
##              -8.487685e-02
##              Garageyes
##              8.003844e-02
##              Lot_Area:MS_ZoningResidential_Low_Density
##              -9.444174e-06
##              Lot_Area:MS_ZoningResidential_Medium_Density
##              5.380959e-07
```

We can read a lot of quantitative relationship from the coefficients above. For example, when holding all other variables fixed, if the overall quality of the house improves by one rating unit, the log of the house price is expected to increase by 0.0888 (equivalently, the house price is expected to increase $(\exp^{0.0888} - 1) * 100\% = 9.29\%$); when the zoning code of the house is **Floating_Village_Residential**, and when the lot area increases by one square foot, the house price is expected to increase by $(\exp^{0.00001246454} - 1) * 100\% = 0.00125\%$.

Open questions / remaining issues: There are six significant outliers (under familywise type I error rate 0.05) in this model:

```
names(which(abs(jack) > cutoff))
```

```
## [1] "126" "710" "754" "2044" "2299" "2893"
```

223 high leverage points also exist. Further validation for these observations based on data sources is desirable. Further improvement may also be made by modeling nonlinearity, such as including polynomials or spline functions of some numerical predictors.